

UI Logical Flow — Qt App and Mobile App

This is not visual UI design.

This is the **screen-to-screen logical flow** for how the Qt app and mobile app should behave.

1. Overall System UI Flow

Start App



Login



Check user role



Open dashboard



User chooses operation:

- Add stock
- Remove stock
- Physical inventory count
- Unknown product review
- Product info management
- Product info suggestion review
- User management
- Sync



Save data locally or officially



Sync through Bluetooth if needed



Qt confirms official data



Transaction history updated

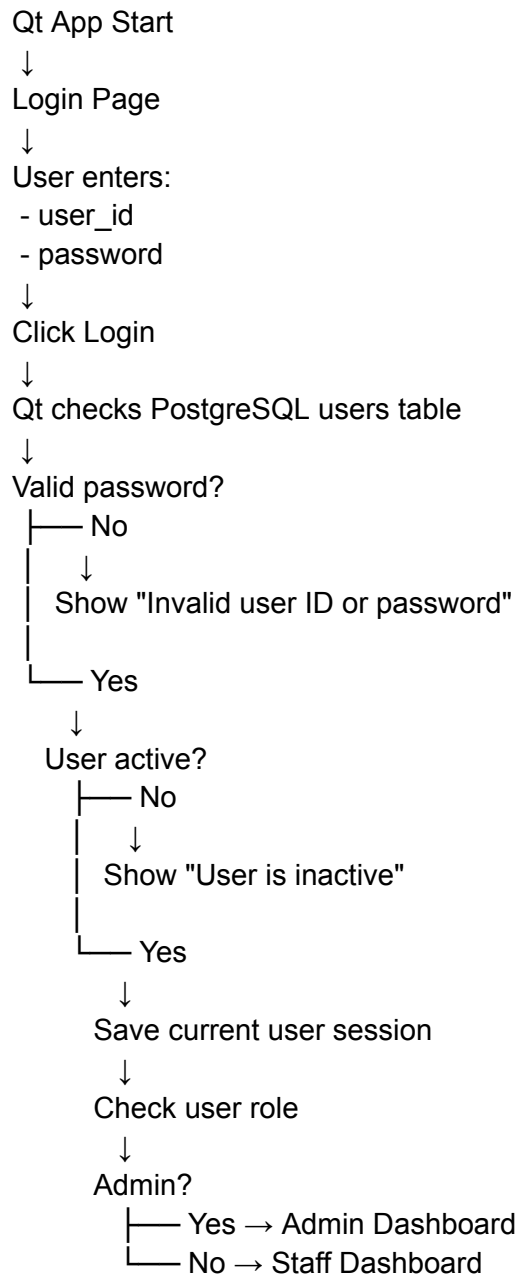
Core rule:

Mobile scans and temporarily stores data.

Qt reviews, confirms, and saves official data.

Part A — Qt App UI Logical Flow

2. Qt App Start Flow



3. Qt Admin Dashboard Flow

Admin should see:

Admin Dashboard

- Product Info
 - Product Info Suggestion Review
 - Unknown Product Review
 - Add Stock Confirmation
 - Remove Stock Confirmation
 - Physical Inventory Count
 - Physical Count Review
 - Stock Quantity
 - Stock Transaction History
 - User Management
 - Sync Status
 - Bluetooth Connection
 - Logout
-

4. Qt Staff Dashboard Flow

Staff should see:

Staff Dashboard

- Add Stock Confirmation
- Remove Stock Confirmation
- Physical Inventory Count
- Physical Count Review
- Stock Quantity
- Stock Transaction History
- Sync Status
- Bluetooth Connection
- Logout

Staff should not see:

User Management

Admin settings

Delete user

Reset other users' passwords

5. Qt User Management Flow

Only admin can access this page.

Admin Dashboard



User Management Page



Show user list:

- user_id
- name
- role
- status
- mobile_number
- last_login_at



Admin chooses:

- Add User
 - Edit User
 - Reset Password
 - Deactivate User
 - Delete User
 - Send User Data to Mobile
-

5.1 Add User Flow

Click Add User



User Create Page



Admin enters:

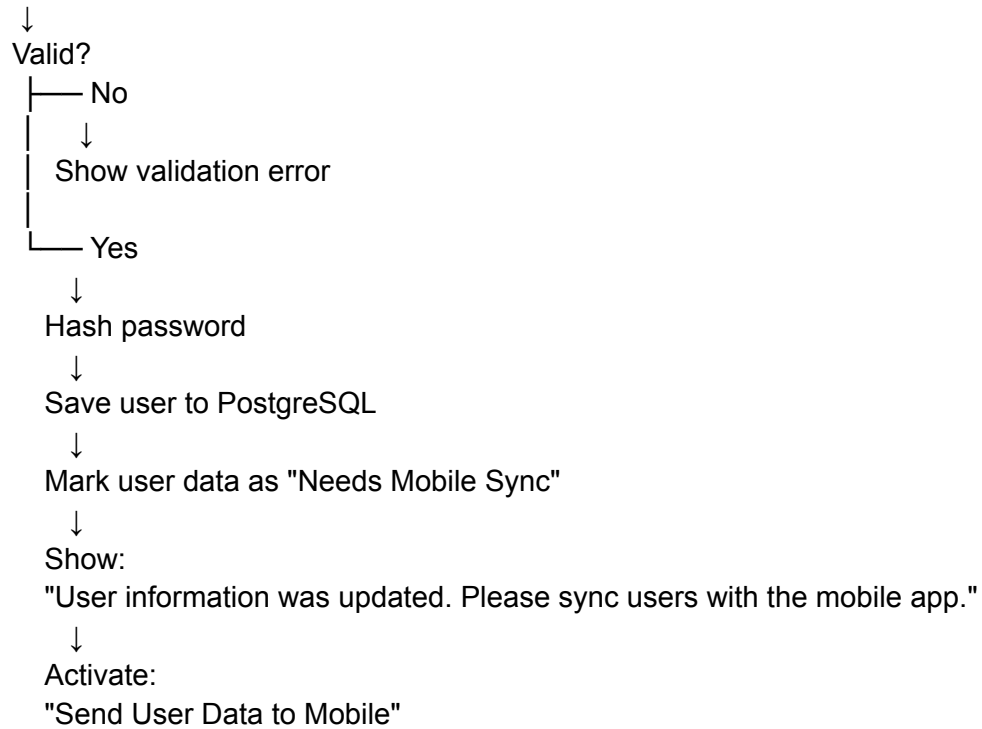
- user_id
- name
- password
- birthday
- mobile_number
- role
- status



Click Save

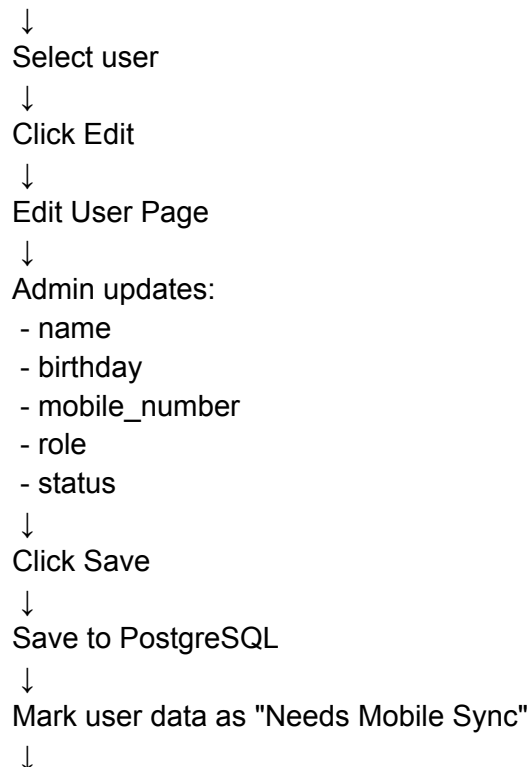


Validate fields



5.2 Edit User Flow

User Management Page



Show sync notification

5.3 Reset Password Flow

Select user



Click Reset Password



Enter new password



Confirm new password



Hash password



Save new password_hash to PostgreSQL



Mark user data as "Needs Mobile Sync"



Show:

"User information was updated. Please sync users with the mobile app."

5.4 Delete or Deactivate User Flow

Select user



Click Delete User



Check if user has transaction history



Has transaction history?

— Yes



Deactivate user instead



Show:

"User has transaction history, so the user was deactivated instead of deleted."



Mark user data as "Needs Mobile Sync"

— No



Delete user or mark as deleted

↓
Mark user data as "Needs Mobile Sync"

6. Qt Bluetooth Connection Flow

Dashboard
↓
Bluetooth Connection Page
↓
Show:
- connection status
- paired mobile device
- last sync time
- pending incoming batches
- pending outgoing batches
↓
Click Connect
↓
Try Bluetooth connection
↓
Connected?
├── No
│ ↓
│ Show "Bluetooth connection failed"
└── Yes
 ↓
 Show "Bluetooth connected"

7. Qt Sync Status Flow

Dashboard
↓
Sync Status Page
↓
Show:
- User data needs sync
- Product info needs sync
- Incoming unknown barcodes
- Incoming stock additions

- Incoming stock removals
 - Incoming physical count sessions
 - Incoming product info suggestions
 - Failed sync batches
 - Duplicate sync batches
-

7.1 Send User Data to Mobile

Click Send User Data to Mobile



Check Bluetooth connection



Bluetooth connected?

— No



Show "Bluetooth not connected"

— Yes



Send user data batch to mobile



Mobile ACK received?

— No



Mark sync failed

— Yes



Mark user sync completed



Show "User sync completed"

7.2 Send Product Info to Mobile

Click Send Product Info to Mobile



Check Bluetooth connection

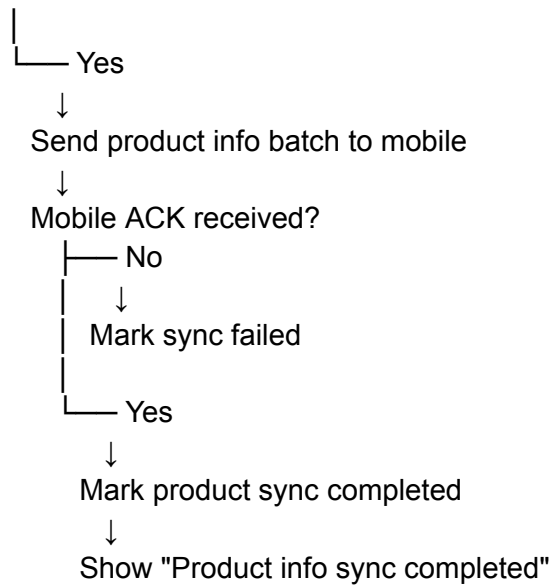


Bluetooth connected?

— No



Show "Bluetooth not connected"



8. Qt Product Info Page Flow

Dashboard



Product Info Page



Show product list:

- barcode
- product_name
- product_type
- default_price
- current_quantity



User can:

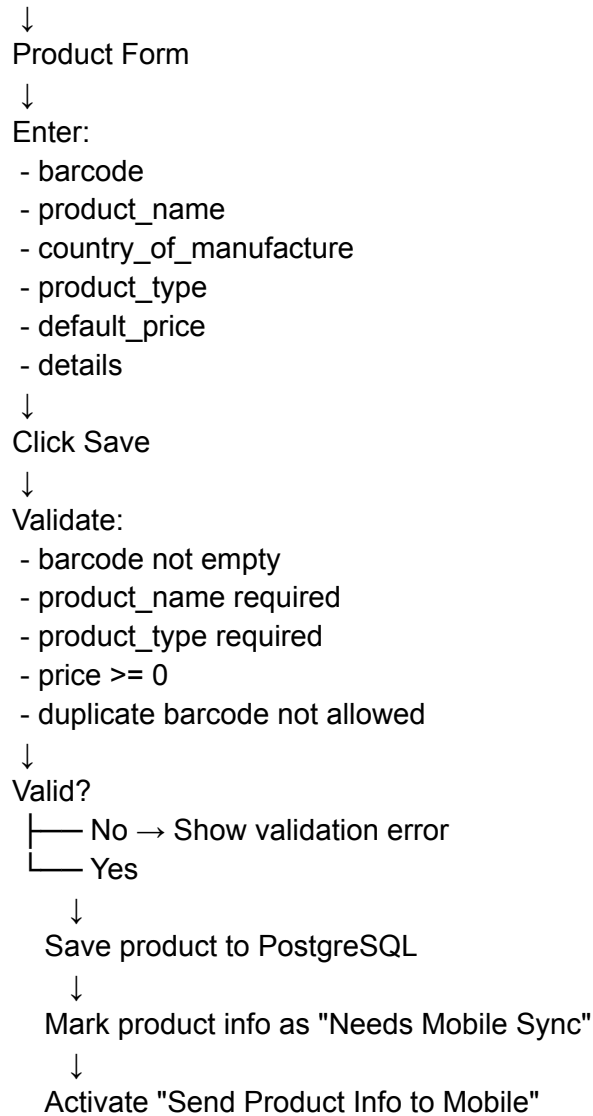
- Search product
 - Add product
 - Edit product
 - View stock history
 - Send Product Info to Mobile
-

8.1 Add Product in Qt

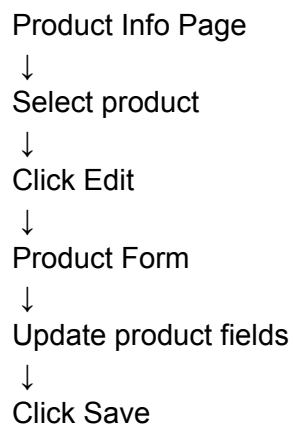
Product Info Page



Click Add Product



8.2 Edit Product in Qt



↓
Validate fields
↓
Save to PostgreSQL
↓
Mark product info as "Needs Mobile Sync"
↓
Show:
"You updated some product information. Please sync it with the mobile app."

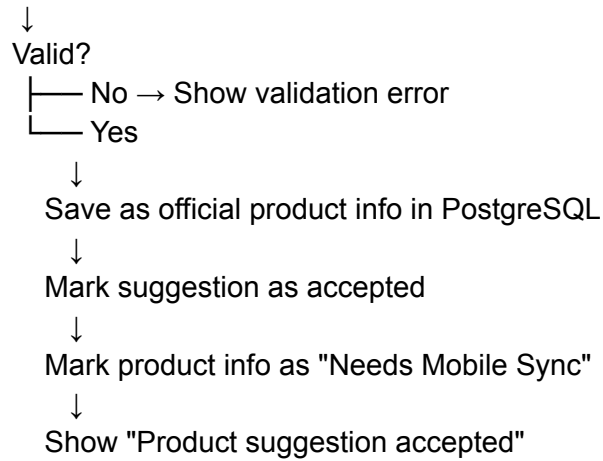
9. Qt Product Info Suggestion Review Flow

This is for product information added or updated on mobile.

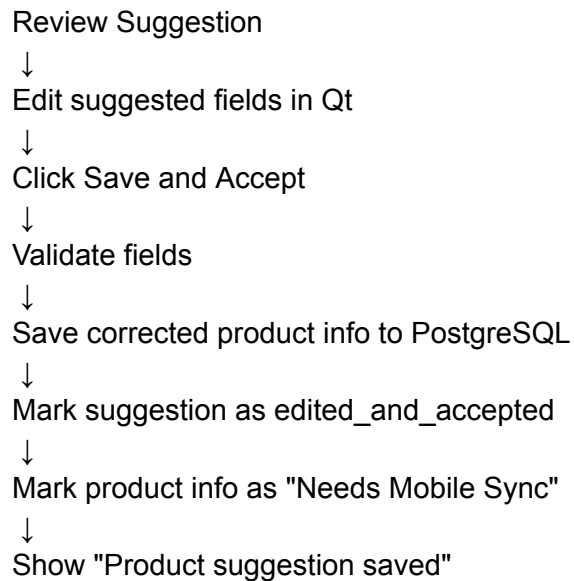
Dashboard
↓
Product Info Suggestion Review Page
↓
Show pending suggestions:
- suggestion_id
- barcode
- suggestion_type: create / update
- suggested_product_name
- suggested_product_type
- suggested_default_price
- created_by_user
- created_at
↓
Select suggestion
↓
Review suggested product info

9.1 Accept Suggestion

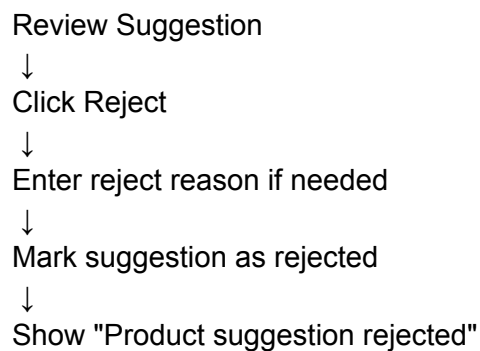
Review Suggestion
↓
Click Accept
↓
Validate product info



9.2 Edit and Accept Suggestion



9.3 Reject Suggestion



10. Qt Unknown Product Review Flow

Unknown product pictures are **not sent to Qt**.

Qt only receives barcode and metadata.

Dashboard



Unknown Product Review Page



Show unresolved unknown items:

- Barcode as main header
- temporary product name if any
- temporary price if any
- pending quantity if any
- first_seen_at
- last_seen_at
- sync_status

Each item header should look like:

Barcode: 888999

The user compares the Qt barcode with the mobile unknown item:

Qt shows barcode.

Mobile shows same barcode + local product picture.

User uses mobile picture as visual reference.

10.1 Resolve Unknown Product

Select unknown barcode in Qt



Look at mobile Unknown List

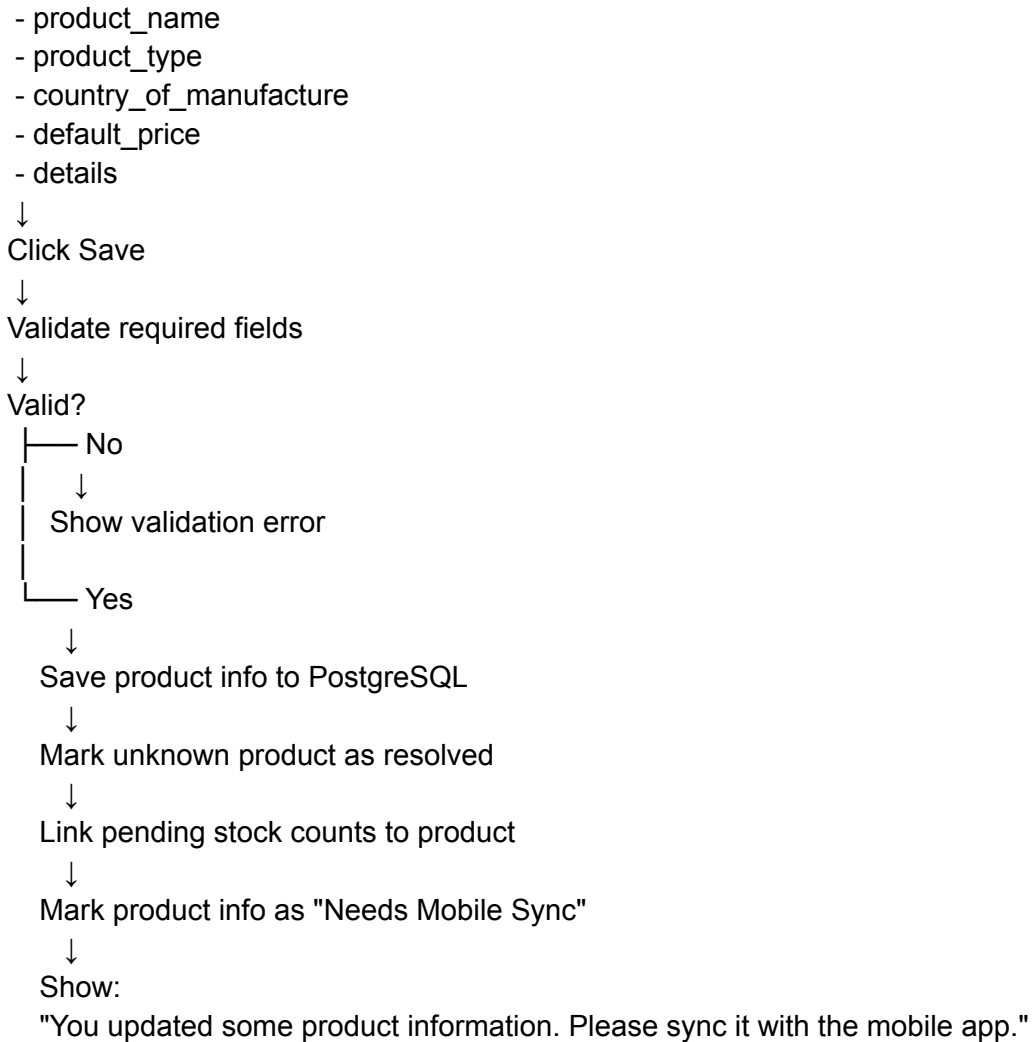


Compare:

- Qt barcode
- Mobile barcode
- Mobile local picture

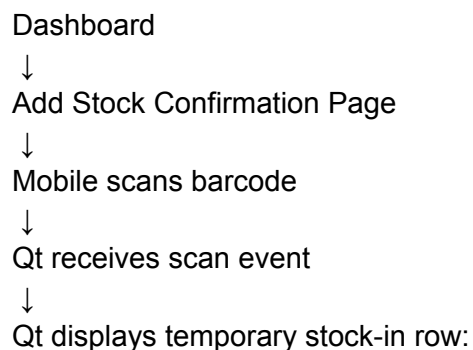


Enter product info in Qt:



11. Qt Add Stock Confirmation Flow

11.1 Real-Time Add Stock from Mobile



- barcode
- product name if known
- temporary quantity
- stock-entry price
- source location
- supplier name
- carried by
- note



User reviews or edits quantity and price



Click "Add Products into Stock"



Validate

Validation:

barcode exists?

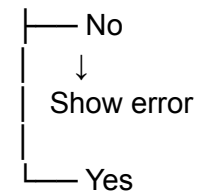
quantity > 0?

price >= 0?

if unknown product, is product info completed?

Flow:

Valid?



Save stock_addition transaction



Increase official stock quantity



Record:

- user_id
- user_name_snapshot
- transaction_time
- device_id



Send ACK to mobile



Reset temporary count



Show "Stock added successfully"

12. Qt Remove Stock Confirmation Flow

Dashboard



Remove Stock Confirmation Page



Mobile scans barcode



Qt receives scan event



Qt displays temporary stock-out row:

- barcode
- product name
- temporary removal quantity
- removal reason
- destination type
- destination name
- destination contact
- destination note



User reviews or edits:

- quantity
- reason
- destination



Click "Remove Products from Stock"



Validate

Validation:

barcode exists?

quantity > 0?

stock quantity enough?

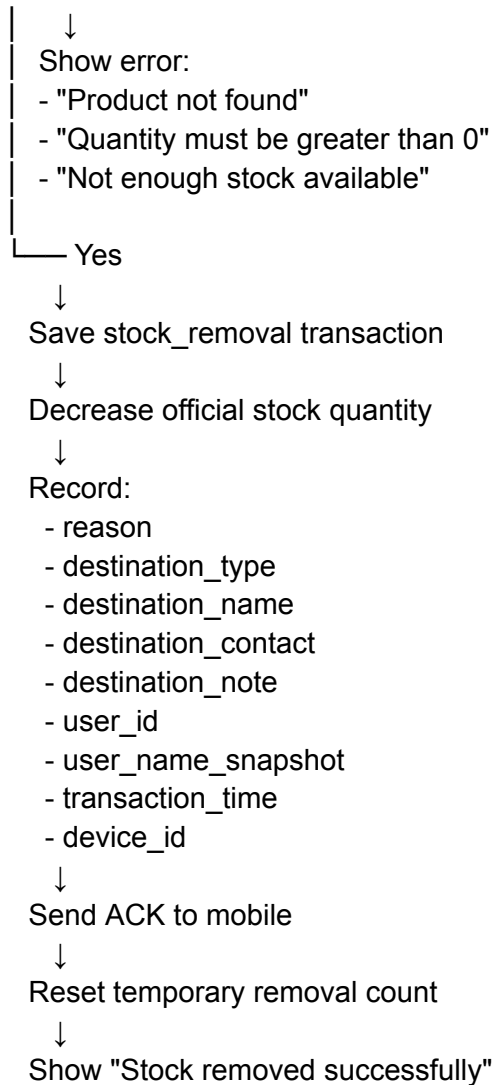
removal reason selected?

destination info valid if entered?

Flow:

Valid?

└─ No



13. Qt Physical Inventory Count Flow

13.1 Start Physical Count

Dashboard



Physical Inventory Count Page



Click Start Count Session



Create count_session_id



Status = counting



Mobile scans products



Qt receives scan events if Bluetooth connected



Qt displays count table:

- barcode
 - product name
 - counted quantity
 - system quantity
 - difference
 - status
-

13.2 Review Physical Count

Click Review Count



Physical Count Review Page



Show each item:

- barcode
- product name
- counted quantity
- system quantity
- difference
- matched / shortage / overage / unknown



Any unknown products?

Yes



Show "Please complete product information first"



Go to Unknown Product Review



No



User confirms adjustment?



No



Save session as reviewed or cancelled



Yes

↓
Create stock_adjustment transactions
↓
Update official stock quantities
↓
Record user_id and transaction_time
↓
Mark count session as applied
↓
Show "Physical count adjustment applied"

14. Qt Stock Quantity Flow

Dashboard

↓
Stock Quantity Page

↓
Show current stock:

- barcode
- product name
- product type
- current quantity
- default price
- estimated value
- last updated time

↓
User can:

- search by barcode
- search by product name
- filter by product type
- open transaction history
- open product detail

15. Qt Transaction History Flow

Dashboard

↓
Stock Transaction History Page

↓
Show transactions:

- transaction_id
- transaction_type
- barcode
- product name
- quantity
- unit_price
- reason
- destination
- user_id
- user_name_snapshot
- transaction_time

↓

User can filter by:

- date
 - user
 - barcode
 - transaction type
 - reason
 - destination
-

Part B — Mobile App UI Logical Flow

16. Mobile App Start Flow

Mobile App Start

↓

Login Page

↓

Choose login method:

- user_id + password
- QR code

↓

Validate using local synced user data

↓

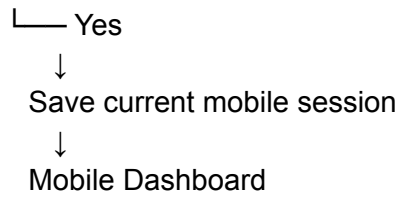
Valid active user?

| — No

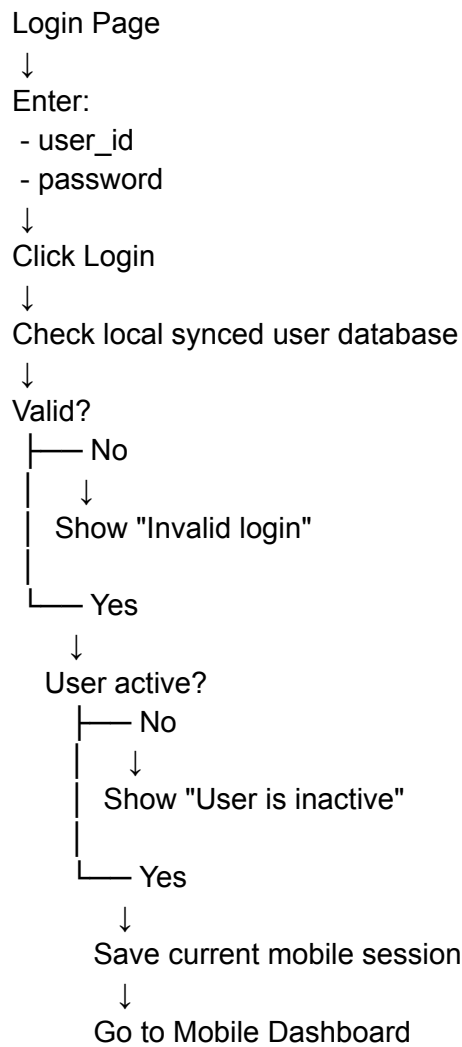
| ↓

| Show login error

|

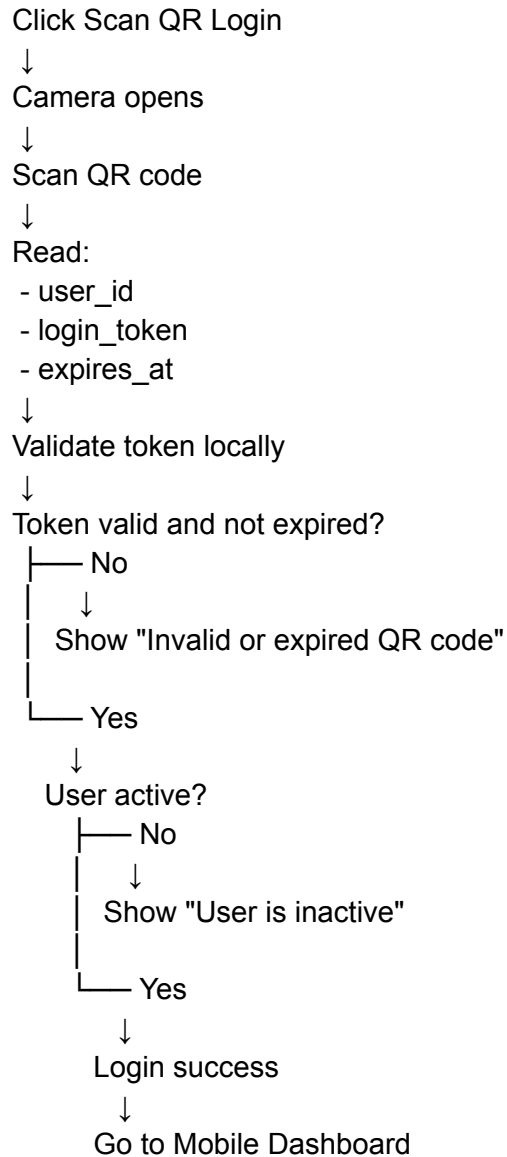


17. Mobile Password Login Flow



18. Mobile QR Login Flow





QR must not contain:

password
password hash
birthday
mobile number

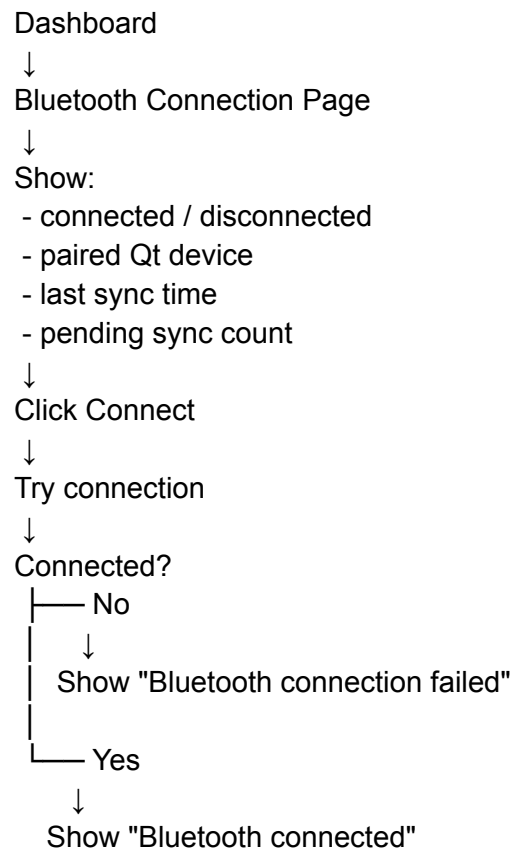
19. Mobile Dashboard Flow

Mobile Dashboard

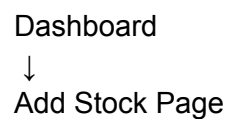
├── Add Stock

- Remove Stock
 - Physical Inventory Count
 - Unknown List
 - Product Info Entry
 - Pending Product Info Suggestions
 - Pending Sync
 - Bluetooth Connection
 - Current User
 - Logout
-

20. Mobile Bluetooth Connection Flow



21. Mobile Add Stock Flow



↓
Scan barcode
↓
Check local product database

21.1 Product Found

Product found locally

↓
Show product info:
- barcode
- product name
- product type
- default price

↓
User enters:
- quantity
- stock-entry price
- source location
- supplier name
- carried by
- note

↓
Bluetooth connected?
├── Yes → Send to Qt
└── No → Save as Pending Sync

21.2 Product Unknown

Product not found locally

↓
Add barcode to Unknown List

↓
Ask:
"Take product picture for local reference?"

↓
User may take picture
↓
Picture is stored only on mobile

↓
Picture is not sent to Qt
↓

User enters:

- quantity
- temporary stock-entry price
- optional temporary product info



Save stock addition as Pending Sync



Send unknown barcode and metadata to Qt when Bluetooth is available

21.3 If Bluetooth Is Connected

Bluetooth connected



Send add-stock data to Qt



Qt ACK received?

— No



Save as Pending Sync



Show "Saved as Pending Sync"



Yes



Show "Sent to Qt"



Final confirmation happens in Qt

21.4 If Bluetooth Is Not Connected

Bluetooth disconnected



Save stock addition as Pending Sync



Show "Stock addition saved for sync"

22. Mobile Remove Stock Flow

Dashboard



Remove Stock Page



Scan barcode



Search local product database

22.1 Product Found

Product found



Show product info:

- barcode
- product name
- product type
- local known quantity if available



User enters:

- removal quantity
- removal reason
- destination type
- destination name
- destination contact
- destination note



Bluetooth connected?

- └─ Yes → Send to Qt
 - └─ No → Save as Pending Sync
-

22.2 Product Unknown

Product not found



Add barcode to Unknown List



Show "Unknown product"



User may take local picture for reference



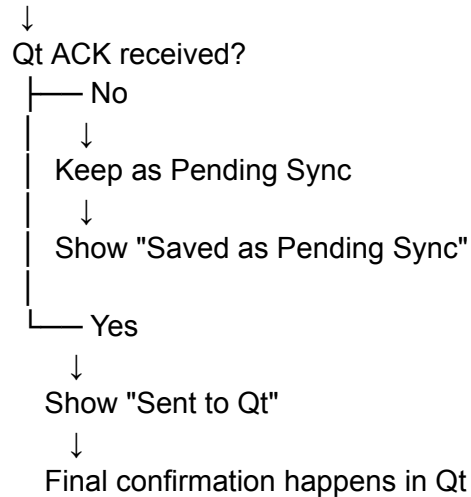
User enters:

- quantity
- removal reason
- destination info

↓
Save stock removal as Pending Sync

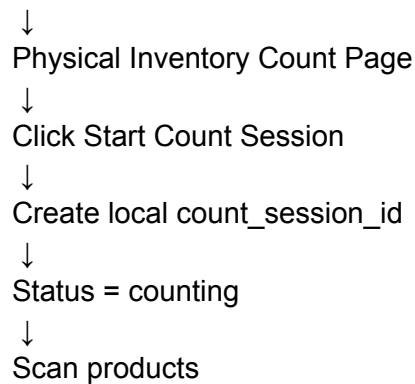
22.3 Send Removal to Qt

Send removal data to Qt



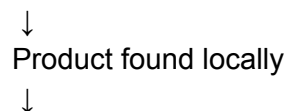
23. Mobile Physical Inventory Count Flow

Dashboard



23.1 Product Found During Count

Scan barcode



Increase counted quantity



Show:

- barcode
 - product name
 - counted quantity
-

23.2 Product Unknown During Count

Scan barcode



Product not found locally



Add barcode to Unknown List



Ask:

"Take product picture for local reference?"



User may take picture



Picture stays only on mobile



Increase counted quantity



Mark item as unknown_product

23.3 Finish Count Session

User clicks Finish Count



Bluetooth connected?

— Yes



Send count session to Qt



Wait for ACK



ACK received?

— Yes → Mark count session as synced

— No → Keep as Pending Sync

— No

↓
Save count session as Pending Sync
↓
Show "Physical count saved for sync"

24. Mobile Unknown List Flow

Dashboard

↓

Unknown List Page

↓

Show unknown products:

- barcode
- local product picture if available
- temporary product name
- temporary price
- pending quantity
- sync status

The picture is only local.

It is not sent to Qt.

24.1 Unknown Product Local Picture Flow

Select unknown barcode

↓

Show:

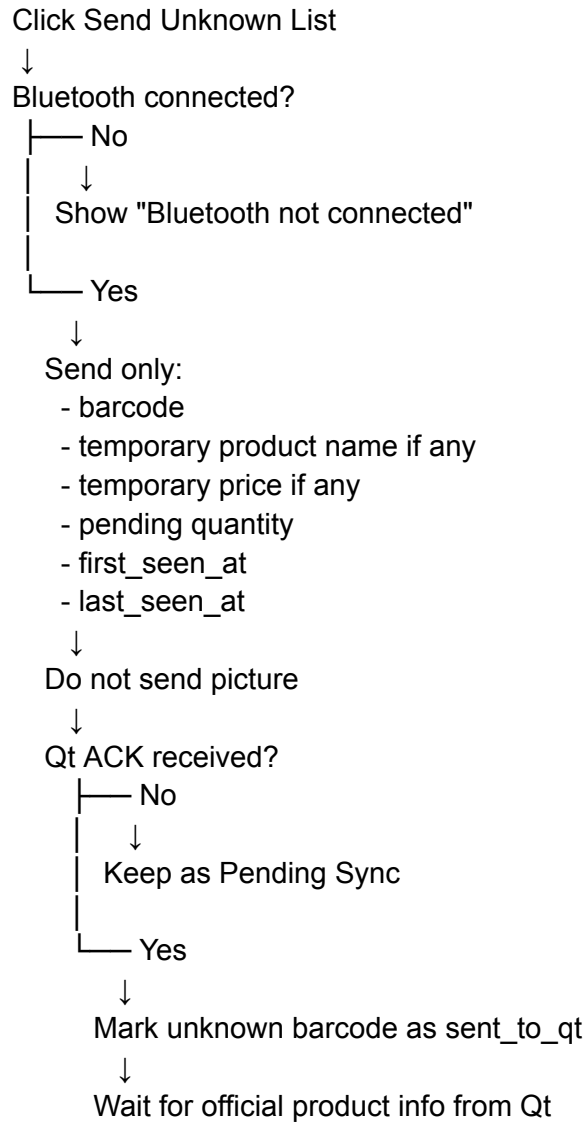
- barcode
- local picture
- temporary fields

↓

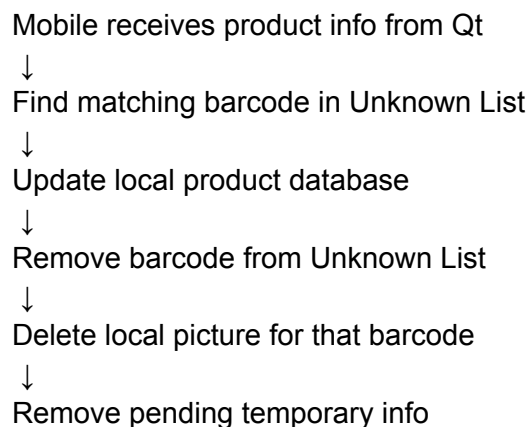
User can:

- take picture
 - replace picture
 - delete picture
 - enter temporary info
-

24.2 Send Unknown List to Qt



24.3 Unknown Product Resolved from Qt



↓
Show "Product information updated"

25. Mobile Product Info Entry Flow

This is for product info added or updated directly on mobile.

Dashboard

↓

Product Info Entry Page

↓

User scans barcode or selects product

↓

Product exists locally?

└─ Yes → Update product info suggestion

└─ No → Create product info suggestion

25.1 Create Product Info Suggestion

Enter:

- barcode
- product_name
- product_type
- country_of_manufacture
- default_price
- details

↓

Click Save

↓

Save locally as:

suggestion_type = create

status = pending_sync

↓

Show:

"Product information suggestion saved."

25.2 Update Product Info Suggestion

Select existing product

↓

Edit:

- product_name
- product_type
- country_of_manufacture
- default_price
- details

↓

Click Save

↓

Save locally as:

suggestion_type = update

status = pending_sync

↓

Show:

"Product information update saved for sync."

25.3 Sync Product Info Suggestion to Qt

Pending Product Info Suggestions Page

↓

Click Sync Suggestions

↓

Bluetooth connected?

└─ No

↓

└─ Show "Bluetooth not connected"

└─ Yes

↓

Send suggestion batch to Qt

↓

Qt ACK received?

└─ No

↓

└─ Keep as Pending Sync

└─ Yes

↓

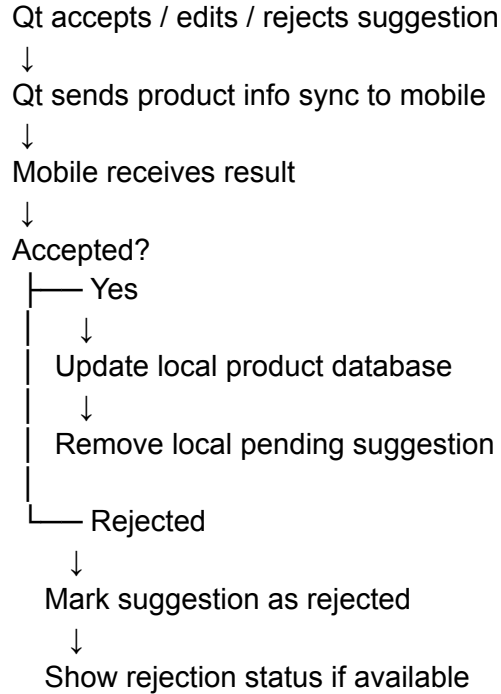
Mark suggestion as sent_to_qt

↓

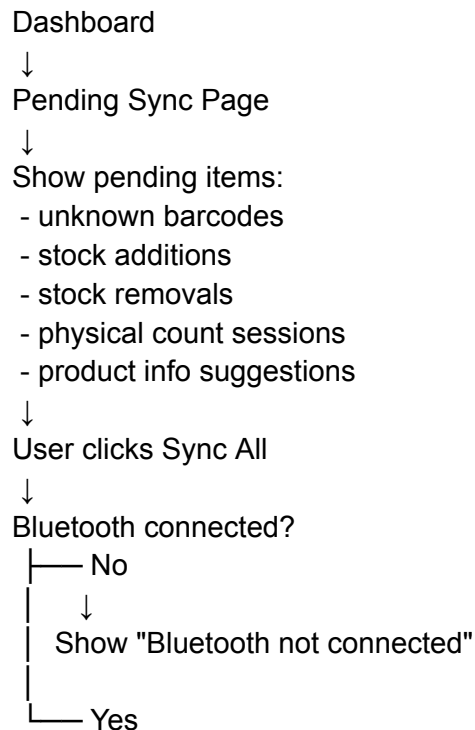
Show:

"Product information suggestion was sent to the Qt app for review."

25.4 After Qt Review



26. Mobile Pending Sync Flow



↓
Send batches one by one

26.1 Batch Sync Rule

For each pending batch:

↓
Send batch to Qt
↓
Wait for ACK
↓
ACK received?
├── Yes
│ ↓
│ Mark batch as synced
└── No
 ↓
 Keep batch as Pending Sync
 ↓
 Show sync failed message

26.2 Duplicate Batch Rule

Send batch to Qt
↓
Qt detects duplicate batch_id
↓
Qt returns duplicate response
↓
Mobile marks batch as already processed
↓
Show:
"This batch was already synced."

27. Mobile Current User Flow

Dashboard
↓
Current User Page

↓

Show:

- user_id
- name
- role
- login method
- last login time

↓

Actions:

- Generate QR Code
 - Logout
-

27.1 Generate QR Code

Click Generate QR Code

↓

Create QR content:

- user_id
- login_token
- expires_at

↓

Display QR code

QR must not include:

password
password hash
birthday
mobile number

Part C — Combined Operation Flows

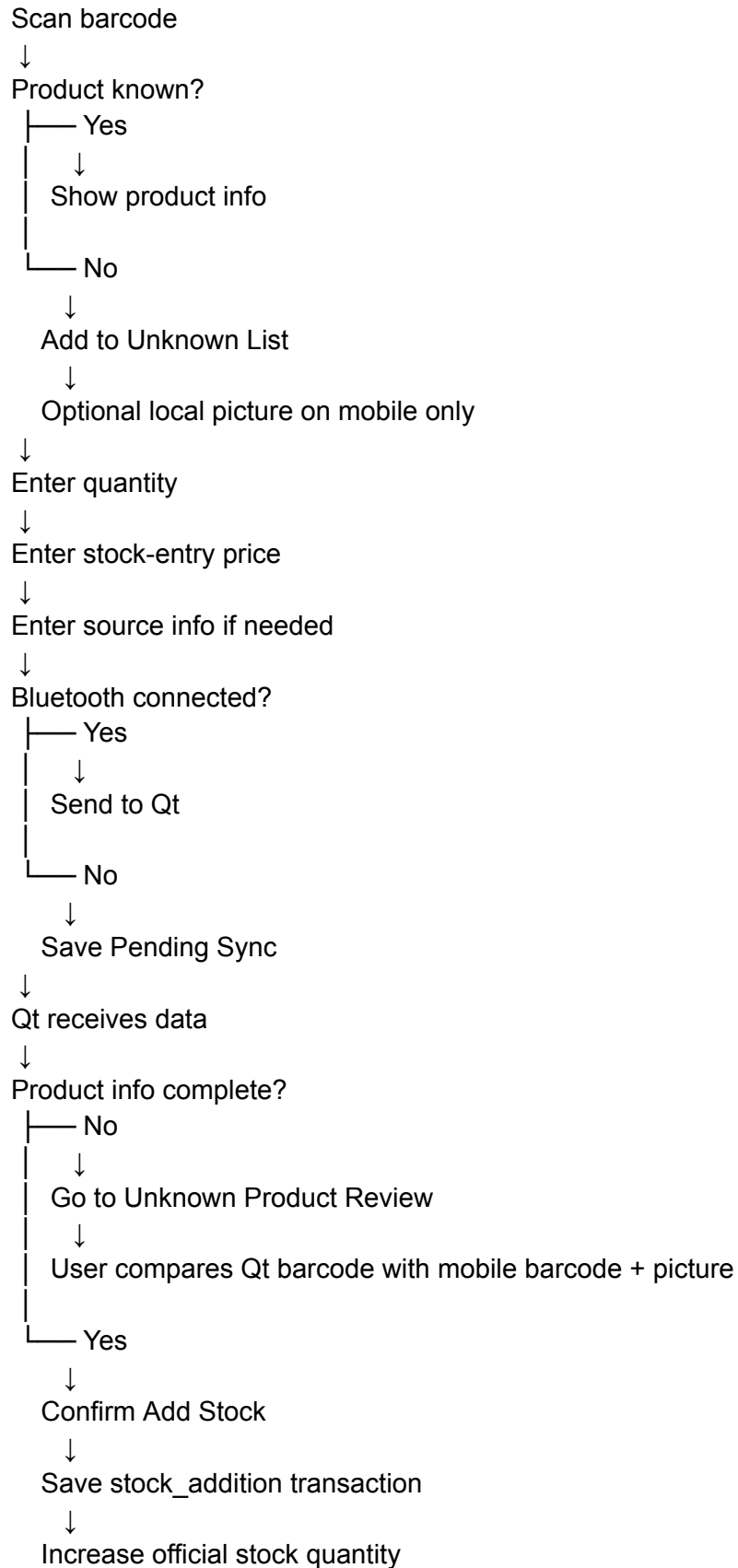
28. Combined Flow — Add Stock

User logs in

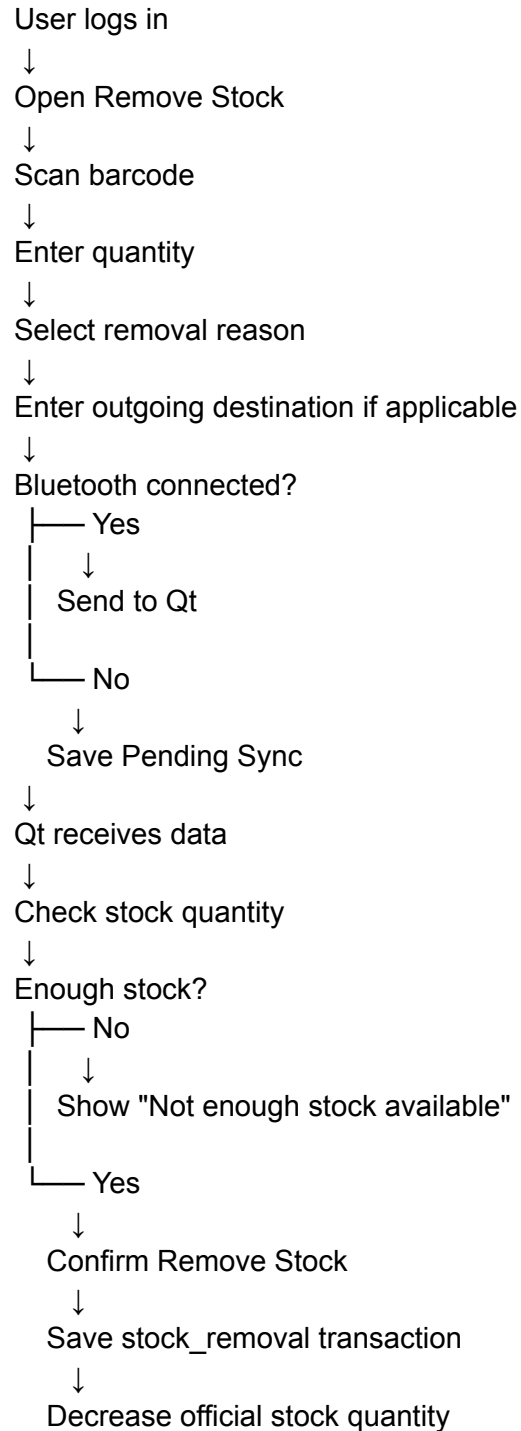
↓

Open Add Stock

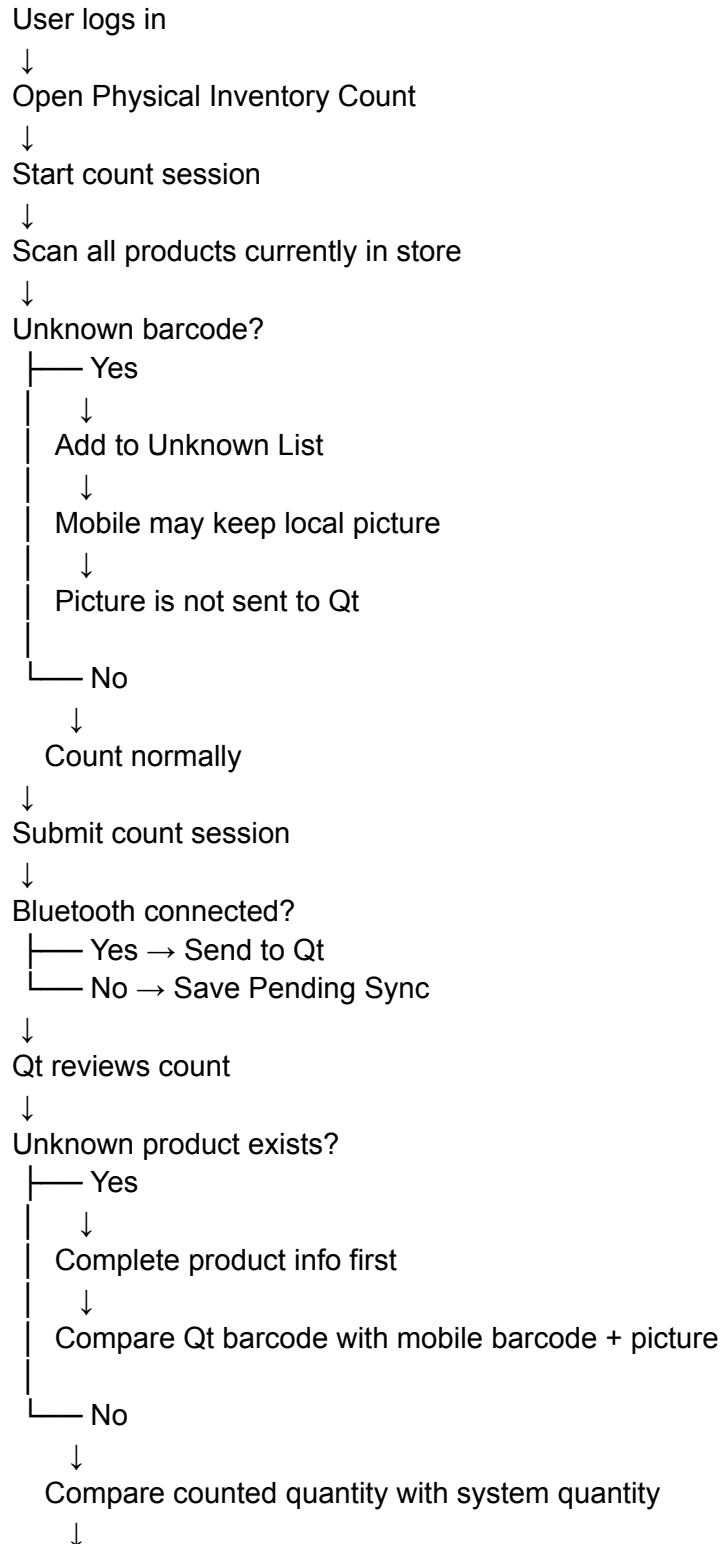
↓

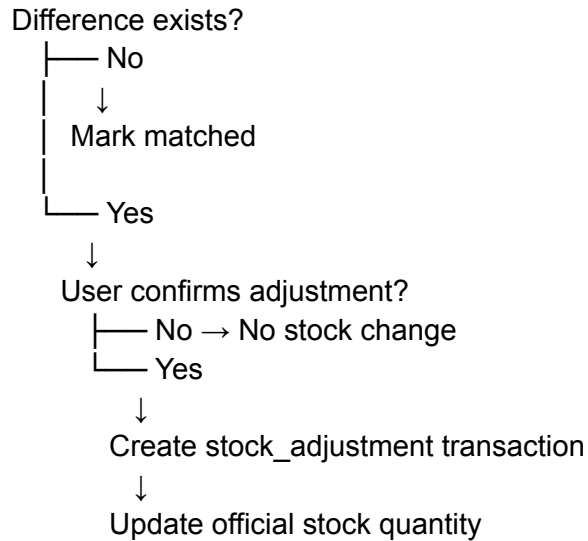


29. Combined Flow — Remove Stock

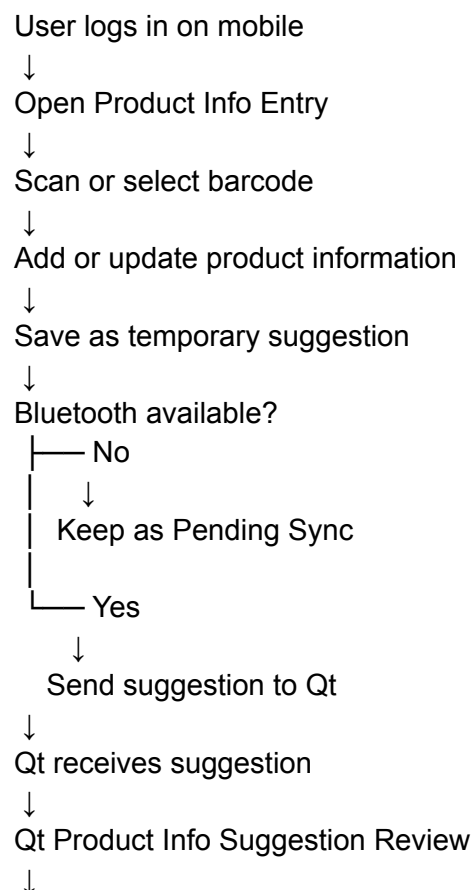


30. Combined Flow — Physical Inventory Count



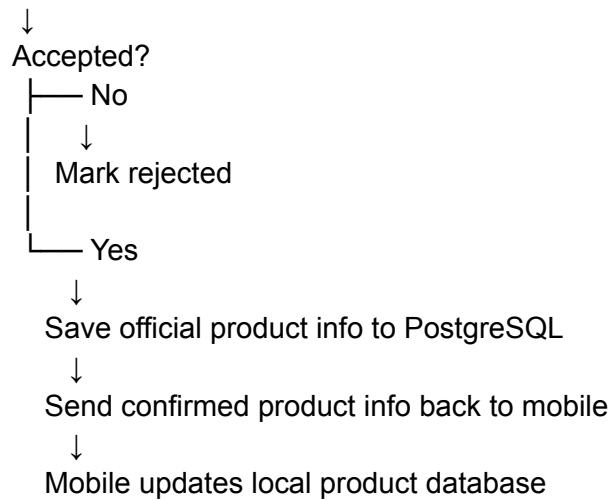


31. Combined Flow — Mobile Product Info Suggestion



Qt user chooses:

- Accept
- Edit and accept
- Reject



32. Recommended UI State Names

32.1 Stock Add State

idle
scanning
counting
pending_confirmation
pending_sync
sent_to_qt
confirmed
failed

32.2 Stock Remove State

idle
scanning
counting
destination_entry
pending_confirmation
pending_sync
sent_to_qt
confirmed

failed

32.3 Unknown Product State

unresolved
picture_local_only
sent_to_qt
resolved_in_qt
synced_to_mobile
cancelled

32.4 Product Info Suggestion State

draft
pending_sync
sent_to_qt
pending_review
accepted
edited_and_accepted
rejected
synced_to_mobile

32.5 Physical Count State

draft
counting
pending_sync
sent_to_qt
reviewed
applied
cancelled

32.6 Sync Batch State

pending
sending
sent
acknowledged
failed
duplicate

33. Most Important UI Labels

The mobile app should clearly show:

Pending Sync
Waiting for Qt confirmation
Sent to Qt
Confirmed by Qt
Local picture only
Product info suggestion pending review

The Qt app should clearly show:

Received from Mobile
Needs Review
Ready to Confirm
Saved to Stock
ACK Sent
Needs Mobile Sync
Suggestion Pending Review

34. Final UI Rule

The UI should always avoid this confusion:

Mobile saved something locally

versus:

Qt confirmed and saved it officially

So the system should use clear status labels:

Local Draft
Pending Sync
Sent to Qt
Needs Qt Review
Confirmed by Qt
Officially Saved